

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 1 SET D

SUMMER SEMESTER, 2024-2025

DURATION: 30 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. Simplify the following boolean expression using cut and try method:

4 + 1
(CO2)
(PO1)

(i) $F_1(P, Q, R) = P\bar{Q}R + \bar{P}\bar{Q}\bar{R} + \bar{P}Q\bar{R} + P\bar{Q}\bar{R} + \bar{P}\bar{Q}R$

(ii) $F_2(P, Q) = (P \oplus Q) + \overline{(P \oplus Q)}$

2. Perform the following subtraction using the $(r-1)$'s complement method without converting the numbers to another base: 4
(CO1)
(PO1)

$$(105)_9 - (472)_9$$

3. Perform the following addition using BCD addition (you have to convert the numbers to BCD, perform the addition, and convert the result back to decimal):
- 5
(CO1)
(PO1)
- $$(689)_{10} + (795)_{10}$$

4. A number $(785)_b$ in an unknown base b is equivalent to the hexadecimal number $(284)_{16}$. Determine the base b .

3
(CO1)
(PO1)

5. Pick the correct option for the following question:

1×3
(CO1,2)
(PO1)

- (A) Can $(0110101)_{GRAY}$ and $(0111101)_{GRAY}$ be two consecutive numbers in the Gray code sequence?

☐ Yes

☐ No

- (B) Does $(1110010)_2$ maintain odd parity?

☐ Yes

☐ No

- (C) Let $F(P, Q, R) = \sum M(1, 3, 4, 6)$. Is $\bar{F} = \sum m(1, 3, 4, 6)$ true?

☐ Yes

☐ No